

Supplementary Methods — StatFootDB non-penalty xG IECV (revised P1+P2+R2)

Supplementary Methods — StatFootDB non-penalty xG internal-external validation

SM0. Literature search and reference verification New references [10]-[24] were screened from a supplied retrieval report for football xG, league-tactical and prediction-methodology relevance. Non-football clinical, EEG, oncology and ICU records from that report were intentionally excluded as football evidence. DOIs and attributions are reproduced as supplied; a final manual verification (exact DOI, correct attribution) is required before submission — see results/REFERENCES_CHECK.csv.

SM1. Reporting and self-audit Reported in line with TRIPOD+AI [9] for prediction-model development and validation. PROBAST+AI [10] used as self-audit grid; signalling questions and judgments are recorded in results/probast_plus_ai_selfaudit.csv (to be deposited). No public preregistration (no OSF/AsPredicted); computational lock with timestamp and hashes in results/computational_lock.json provides retrospective internal locking only.

SM2. Cohort audit detail (S1) Supplementary Table S1 (results/supplementary_S1_audit.csv) gives match retention by league-season from audit/primary_candidate_retention_by_stratum.csv. Global flag counts from audit/match_quarantine_flags.json: team_shot_count_difference 264, total_shot_count_difference 236, score_or_goal_event_difference 31, team_score_difference 31, unresolved_shooter_team 29, invalid_date 5, unresolved_identical_shot_pair 1, outside_regular_season_round_range 1; union 277 unique matches (flags overlap, do not add). Sequential shot accounting is Table A1. The 277-match union (match unit) and the 5,944-shot final restriction (shot unit, Table A1) must not be added. Detailed per-flag by league-season breakdown is in audit/match_quarantine_flags.json and audit/shot_counts_against_team_stats.json.

SM3. Geometry and orientation checks Reference pitch 105 by 68; attacked goal at (0,34); posts at (0,30.34) and (0,37.66); goal width 7.32 as geometric assumption. Distance $\sqrt{x^2+(y-34)^2}$; angle $\text{atan2}(|u_x*v_y-u_y*v_x|, u.v)$. Implementations cross-checked by cosine-law and difference-of-bearings on all 214,143 primary shots. Endpoint rule $x_2 > x_1$ disagreed with team-based orientation on 1,039 raw records (776 own goals + 263 blocked); blocked

shots retained. Internal agreement does not establish broadcast-level accuracy.

SM4. Model fitting and tie-breaking 24 primary base fits + 4 temporal + 6 broad-cohort fits + development-fold fits + post-hoc calibration. Candidate order broke exact numerical ties; no final-test criterion used. Common M3: `learning_rate` 0.05, `min_samples_leaf` 100, `l2` 1, `early_stopping` False. Narrow grids C in {0.1,1}, leaves in {7,15} as deliberate anti-overfitting constraint on two chronological folds.

SM5. Verification tolerances Independent recomputation over 156 metric rows; largest absolute discrepancy for log-loss/Brier/rank-AUC below $9e-15$; 24 base models reloaded, 2,880 sampled predictions recomputed; training-only medians, prohibited predictors, outcome alignment, macro averages and primary effect checked; 20 source-file hashes and protocol hash unchanged. Full logs in `results/independent_results_verification.json` and `results/third_numerical_check.json`.

SM6a. Worked heterogeneity example (Bundesliga, descriptive DerSimonian-Laird) League SEs from 95% interval widths/3.92: 0.000207, 0.000298, 0.000132 (Bundesliga), 0.000238, 0.000225. Fixed-effect mean -0.000029 gives $Q=8.268$ on 4 df; with $C=\text{sum}(w)-(\text{sum}(w^2)/\text{sum}(w))$ over fixed-effect weights, $\text{tau-squared}=\max(0,(Q-\text{df})/C)=4.58e-08$ (classical DL method-of-moments). Random-effects variance for Bundesliga = $\text{SE}^2+\text{tau}^2 = 1.7424e-08+4.578e-08 = 6.3204e-08$, $\text{weight} = 1/6.3204e-08 = 15,821,783$ (28.9% of total weight 54,671,319). Weighted mean over the five weights = -0.000087 (SE 0.000135). The 95% prediction interval uses a t multiplier with $k-2=3$ df: $-0.000087 \pm t(0.975,3)*\sqrt{\text{tau}^2+\text{SE_RE}^2} = [-0.000893,0.000719]$. This worked example is part of the independent recomputation checks (Section 2.6).

SM6b. Exploratory meta-regression (league effect on feature shift, $n=5$) Ordinary least squares of the five league B2-B0 point estimates on calibration-to-test Wasserstein distance for distance gave a slope of +0.00024 per metre (R-squared 0.04, $n=5$); on Jensen-Shannon divergence (base 2) for situation, R-squared was 0.27. With five overlapping-training estimates, neither permits conclusions; the analysis is reported so that Sections 3.5 and 3.7 are linked rather than leaving feature shift purely descriptive. Full inputs are in `results/supplementary_S3_distances.csv` and `results/bootstrap_B2_vs_B0.csv`.

SM5b. Feature-shift distance definitions (Table S3b) For continuous features (distance, angle, minute), Wasserstein-1 distance $W(P,Q)=\text{integral}|F_P-F_Q|$

between calibration and test empirical distributions (`scipy.stats.wasserstein_distance`). Standardized mean difference $SMD = (\text{mean_test} - \text{mean_cal}) / \sqrt{(\text{var_test} + \text{var_cal}) / 2}$. For categorical features (body part, situation), Jensen-Shannon divergence (base 2) between calibration and test category proportions (`scipy.spatial.distance.jensenshannon`). These are descriptive transport diagnostics, not evidence of conditional probability drift.

SM6. Additional bootstrap specifications Primary: 2,000 paired match-cluster replicates, percentile 2.5/97.5 (not BCa), clusters 375/296/295/371/368 per league. Calendar-week blocks and club-season clusters (18-20 teams/league, 2,000 replicates) as sensitivities. Calibration (a,b) CIs: 400 calibration-season bootstrap replicates with L-BFGS-B refits, seed 20260910 (`results/supplementary_S2_calibration_params_CI.csv`). AUC CIs: 2,000 match-cluster replicates on B/M3 test predictions (`results/supplementary_AUC_club.csv`). Heterogeneity: DerSimonian-Laird τ^2 , I², RE mean and t-based 95% PI with 3 df (results in Section 3.7; script `tmp/opencode/calc_remain1.py` logic retained in reproducibility package).